Progressive Recurrent Neural Network for Multispectral Remote Sensing Image Destriping

Jia Li, Junjie Zhang, Jungong Han, Chenggang Yan, and Dan Zeng, *Senior Member, IEEE*

***Abstract*— An unstable imaging system often introduces additional stripe noise in multispectral remote sensing images during the data acquisition process given a variety of factors. The complicated stripe distributions lead to the residual stripe in the results of existing methods, thus increasing the difficulty of destriping in practice. Mainstream deep-learning-based methods show the encouraging destriping performance on multispectral remote sensing images. However, they often require the model to handle the varying degrees of stripe noise in a single shot for each image, which results in the poor destriping performance when facing practical cases with diverse stripe distributions. To address the above issue, we propose a progressive recurrent neural network (PRNet) to remove the stripe noise for each degraded image in an iterative manner. More specifically, a progressive destriping strategy is designed to gradually restore the clean image, in which the main recurrent module (MRM) is introduced to iteratively process the stripe removal results generated from previous timesteps until the clean image is obtained. Furthermore, since the uniformity of the entire image is supposed to be significantly enhanced after destriping, it is necessary to take the local spatial correlation into account during destriping. Therefore, we present the patch-based sequence module (PSM) to leverage the local spatial correlation by splitting the image into multiscale patch sequences and capturing the relationship among different patches. Extensive experimental results on different datasets demonstrate that the proposed model yields superior destriping performance compared with other methods, especially for removing the stripe noise with complex distributions.**

***Index Terms*— Convolutional long short-term memory (ConvLSTM), destriping, multispectral remote sensing image, progressive destriping strategy.**

Manuscript received 27 June 2023; revised 15 September 2023; accepted 8 October 2023. Date of publication 13 October 2023; date of current version 25 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62372284 and Grant 62202283. *(Jia Li and Junjie Zhang are co-first authors.) (Corresponding author: Dan Zeng.)*

Jia Li is with the National Engineering Laboratory for Internet Medical Systems and Applications, The First Affiliated Hospital of Zhengzhou University, Zhengzhou 450052, China, and also with the Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Joint International Research Laboratory of Specialty Fiber Optics and Advanced Communication, Shanghai Institute of Advanced Communication and Data Science, Shanghai University, Shanghai 200444, China (e-mail: 13676952257@163.com).

Junjie Zhang and Dan Zeng are with the Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Joint International Research Laboratory of Specialty Fiber Optics and Advanced Communication, Shanghai Institute of Advanced Communication and Data Science, Shanghai University, Shanghai 200444, China (e-mail: junjie_zhang@shu.edu.cn; dzeng@shu.edu.cn).

Jungong Han is with the Department of Computer Science, University of Sheffield, S1 4DP Sheffield, U.K. (e-mail: juh22@aber.ac.uk).

Chenggang Yan is with the School of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310018, China, and also with the Lishui Institute, Hangzhou Dianzi University, Lishui 323000, China (e-mail: cgyan@hdu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3324606

I. Introduction

MULTISPECTRAL remote sensing images are valuable resources due to their abundant spatial and spectral information. However, the output from adjacent detectors may vary in digital numbers even when imaging the same ground material due to the interference of various factors in the unstable imaging system. This inconsistency often results in stripe noise in the original images, which significantly degrades the quality of data and compromises the performance of downstream tasks, such as endmember extraction [1], unmixing [2], classification [3], segmentation [4], and target detection [5]. Consequently, destriping plays an essential role in remote sensing image processing to improve the data quality and promote the subsequent applications.

To address the issue of stripe noise in multispectral remote sensing images, destriping methods have been developed. The existing destriping methods can be categorized into four types: statistical-matching-based [6], filtering-based [7], optimization-based [8], [9], and deep-learning-based [10]. However, traditional methods often have limitations, such as ineffective performance for complex ground materials, unsuitability for images with significant characteristics in the transformed domain, and invalidation of the stripe removal due to rigid assumptions. Deep-learning-based methods have shown promising destriping performance in recent years, leveraging the representation capability of convolutional neural networks (CNNs). These methods design a variety of network structures to improve the learning ability of the model. For instance, Xiao et al. [11] introduced the residual learning strategy to the destriping for infrared cloud images (ICSRN). To better use the spectral correlation and the structural feature, a spatial–spectral gradient network (SSGN) is proposed by Zhang et al. [12] for hyperspectral image denoising. However, a common issue with these methods is that they aim to eliminate the stripe in a single shot, which may result in residual stripes in the corrected results when dealing with complicated and severe stripe noise.

To address this issue, we propose a progressive recurrent neural network (PRNet) that removes the stripe noise iteratively for each degraded image. The proposed PRNet uses a progressive destriping strategy that gradually restores the clean image by removing the stripe with multiple timesteps. We consider the processed results of adjacent timesteps as a sequence and fully explore their relevance during the iterative destriping. To exploit the sequential information, we propose

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

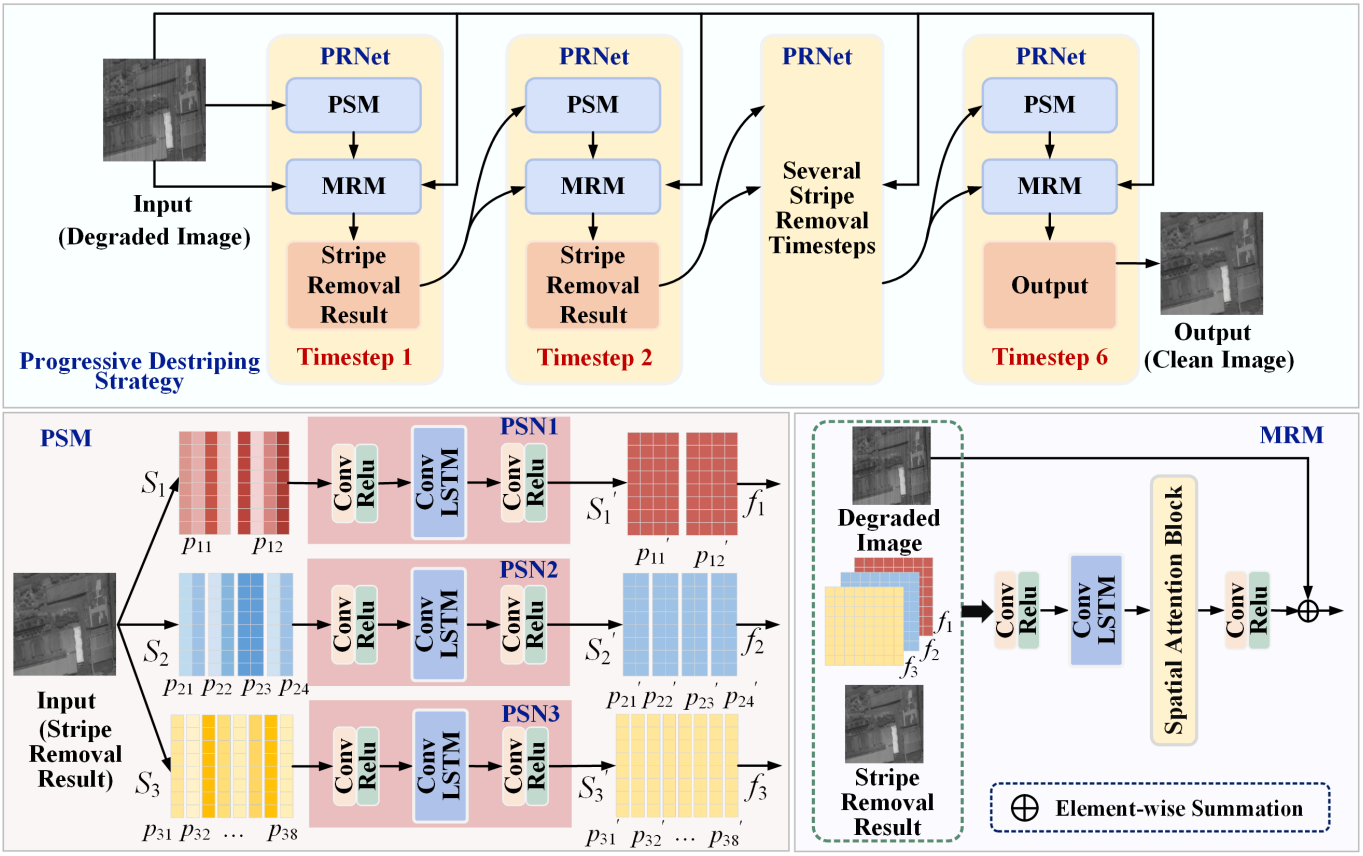


Fig. 1. Structure of the proposed PRNet. In the progressive destriping strategy, the model removes the stripe in the input degraded image in multiple timesteps until the clean image is obtained. The PRNet contains two modules, namely, PSM and MRM. The PSM splits the stripe removal results generated by the previous timestep into several patches with three scales and explores the local correlation in three patch sequences. The MRM iteratively implements destriping in multiple timesteps, while the correlation between stripe removal results is learned.

a main recurrent module (MRM) based on the convolutional long-short term memory (ConvLSTM) [13].

In addition, the existing methods generally use spatial information to promote destriping performance, but the local spatial correlation has yet to be explored. For instance, Chang et al. [14] presented an anisotropic spectral–spatial total variation (ASSTV) model for destriping, in which the spatial smoothness is enhanced. The horizontal and vertical gradients of the degraded images are fed to the network in the deep-learning-based method SSGN [12]. In the above methods, the total variation (TV) model aims at reducing the difference between adjacent columns, while the SSGN obtains the horizontal and vertical gradients by calculating the differences between the adjacent columns and rows. It can be observed that these methods simply use the local spatial information of the image, and the local spatial correlation is yet to be explored. Since the uniformity of the entire image is expected to be reinforced by the elimination of the stripe noise, we consider the local spatial correlation as an essential factor in maintaining the overall structural information of the clean image. To maintain the overall structural information, we introduce the patch-based sequence module (PSM) to explore the local spatial correlation. The PSM is shown in the left part of Fig. 1. Since the size of patches affects the uniformity of the entire image, we use an overlapping strategy to sample the patch sequences. The ConvLSTM is then used to capture the correlation between the adjacent patches in different sequences.

Overall, the major contributions can be summarized as follows.

1) This article proposes a novel destriping method, the PRNet, to remove stripe noise in multispectral remote sensing images. The proposed PRNet uses a progressive destriping strategy that gradually restores the clean image in an iterative manner, enabling the method to handle diverse degrees of stripe noise. To the best of our knowledge, this is one of the first attempts to address multispectral remote sensing image destriping in an incremental manner.
2) To capture the relevance of detailed image information among the processed results of adjacent timesteps, we present the MRM, which implements stripe removal in multiple timesteps. The MRM enhances the learning ability of the model by exploiting sequential information and effectively removing stripe noise.
3) Moreover, to strengthen the uniformity of the entire image, we introduce the PSM to exploit the local spatial correlation. The proposed PSM splits the image into multiscale patch sequences along the column direction, given the unique structural characteristic of stripe noise.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

The PSM effectively removes stripe noise by capturing the correlation between adjacent patches in different sequences.

II. Related Works

A. Statistical-Matching-Based Destriping Methods

Assuming that the imaging system acquires enough historical data, each column of pixels in the image thus has the same statistics. Following the above observation, the statistical-matching-based methods adjust the data in each column by calculating the statistics to accomplish stripe removal. They are generally composed of two stages: clean reference allocation and handcrafted statistical matching.

Histogram matching and moment matching (MM) are representatives of the statistical-matching-based methods. Horn and Woodham [15] adjusted the histogram of each column to a clean reference to eliminate the stripe for the Landsat image. However, the histogram statistics of each column are different when the data are insufficient, resulting in poor destriping performance. Thus, Wegener [16] overcame these drawbacks by assigning homogeneous regions to optimize the search strategy of the reference. Gadallah et al. [17] used the moment features, which is more robust compared with the histogram matching relying on similarity assumption. Although the statistical-matching-based destriping methods are efficient to implement, such strong assumption may become invalid with the complex ground object distribution in practice. This issue severely limits the effectiveness of these methods for destriping. Moreover, they tend to generate unsatisfactory results in some particular cases, such as the broken stripe that incompletely covers the whole column, resulting from incorrect matching information.

B. Filtering-Based Destriping Methods

Since the features of different image constituents are more obvious in the transformed domain compared with its original counterpart, filtering-based methods transfer the degraded images into the transformed domain to eliminate the stripe component. If the stripe noise displays a strong periodicity in the original image domain, a suitable filter in the frequency domain is designed by the Fourier transform methods for destriping. The wavelet decomposition methods extract the information of images using the basic wavelet function, while the direction of the stripe in the remote sensing image is taken into account.

Simpson et al. [18] introduced a finite-impulse response (FIR) filter to remove the stripe by following three steps: image transformation based on fast Fourier transformation (FFT), stripe characteristic analysis, and filter design for destriping. Similarly, Chen et al. [19] also proposed an FIR filter for Chinese moderate imaging spectroradiometer (CMODIS) images' destriping. Moreover, a method based on wavelet analysis for hyperspectral image destriping was developed by Pande-Chhetri and Abd-Elrahman [20], while the adaptive Fourier zero-frequency amplitude normalization is used to promote the performance. However, since the stripe and clean components are not always correctly separated in the transform domain, there is a loss of useful signal during destriping based on the Fourier transform. In addition, sudden changes in the input radiance can lead to the ringing artifacts in the restored results [21]. Subsequently, Münch et al. [22] developed the Fourier transform methods through using the wavelet decomposition, which considers the scaling directional properties and effectively promotes the destriping performance. A limiting factor of such methods is that a correct wavelet transform function needs to be chosen to guarantee a suitable transformation of the data. Furthermore, some destriping methods that fuse statistical matching and filtering have been proposed [23], [24], [25], [26]. Generally speaking, satisfactory destriping performance is achieved by the filtering-based methods when handling the stripes with strong periodicity, and the effectiveness can be limited, given the real stripe possesses complex characteristics.

C. Optimization-Based Destriping Methods

The optimization-based methods view stripe removal as an ill-posed inverse problem. Objective function designing and stripe removal are two crucial steps in these methods. As expressed in the following equation, the objective function typically composes of the data-fidelity term $\|\mathbf{H}-\mathbf{X}-\mathbf{S}\|_F^2$ and the prior focusing on the characteristic of the image $\mathbf{X}$ or stripe $\mathbf{S}$:

$$\{\hat{\mathbf{X}},\hat{\mathbf{S}}\}=\arg\min_{\mathbf{X},\mathbf{S}}\frac{1}{2}\|\mathbf{H}-\mathbf{X}-\mathbf{S}\|_F^2+\lambda\phi(\mathbf{X})+\rho\psi(\mathbf{S}). \quad (1)$$

According to the way data are processed, there are three categories among these methods, i.e., variation, low-rank (LR) matrix, and LR tensor.

The first one generates a clean image based on the variation model considering the prior knowledge. Shen and Zhang [27] first adopted the Huber–Markov-based variational model to optimize the objective such that the clean image is restored. This method effectively preserves some detailed information, such as the edge of the image, while the directional structure of the stripe is ignored. Therefore, Bouali and Ladjal [28] introduced a delicate unidirectional variational model to overcome the above shortcoming. Specifically, the structural information of the original image is preserved to a large extent when the stripe noise is removed by minimizing the horizontal gradients and retaining the vertical ones. Moreover, Chang et al. [14] improved the unidirectional variational model to enhance the image in terms of smoothing from different perspectives such as spectral and spatial. There are also destriping methods using the unidirectional TV model [29], [30]. In addition, since the stripe component possesses prominent structural characteristics compared with the clean image, directly reducing the noise component is also a feasible solution [31], [32].

For the hyperspectral image, the LR property is an essential prerequisite to constrain the optimization goals. The LR matrix-based methods transfer the 3-D cubic into a 2-D matrix in lexicographical order by considering the spectral coherence [33]. For example, Lu et al. [34] ordered the hyperspectral data into a 2-D matrix, and subsequently, a robust principal component analysis (RPCA) [35] was adopted to decompose

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

the degraded image so that the sparse stripe noise is obtained. Different from applying the LR constraint on the image, Chang et al. [36] and [37] pointed out that it is more efficient to apply the LR constraint to stripe components than images in the estimation of the expected results. Furthermore, the local or nonlocal sparsity is considered to promote the estimated results for the LR model [34], [38], [39], [40], [41].

Although the matrix-based methods achieve promising destriping performance, the losses of spectral–spatial structural correlation are inevitable due to the matrix technique to 3-D data. This issue is alleviated with the emergence of LR tensor-based destriping methods. Xie et al. [42] proposed a tensor-based method by leveraging two intrinsic characteristics, including the global spectral correlation and the nonlocal spatial self-similarity. Similarly, Chen et al. [43] designed a decomposition framework combined with the LR property to remove the stripe by decomposing the image and stripe component. Fan et al. [44] introduced a method that incorporates TV regulation and LR tensor factorization to remove the mixed noise for the hyperspectral image. However, these methods simply use priors to each tensor model. Therefore, to model the intrinsic correlation of different tensor modes, Chang et al. [45] proposed a unidirectional LR tensor recovery model and used the analysis-based hyper-Laplacian priors to model global spectral structures. Although optimization-based methods achieve advanced destriping performance compared with the previous ones, their speed is relatively slow given the complex computation for the large data size. Consequently, this kind of method cannot be applied to cases requiring real-time performance.

D. Deep-Learning-Based Destriping Methods

Most methods described above make different assumptions or use certain priors to promote the destriping effect. However, the effectiveness of these methods is somewhat limited when dealing with real images due to idealized assumptions or priors. In recent years, inspired by the success of deep learning on other tasks [46], [47], [48], deep CNN-based models have been implemented for the image destriping task without the handcrafted assumptions and priors, becoming the most advanced methods [49]. The supervised CNNs can handle the arbitrarily mixed stripe relying on their powerful feature extraction ability.

Kuang et al. [50] designed a three-layer CNN for nonuniform correction to infrared images. Subsequently, the residual learning strategy was introduced to destriping for infrared cloud images by Xiao et al. [11], in which the local–global combination structure is used to recover the details. Zhang et al. [12] constructed the SSGN to use the multiscale features to capture sufficient image information, such that the different kinds of noise, including Gaussian, stripe, and so on, can be reduced. Moreover, Chang et al. [33] focused on the directional characteristic of the stripe and designed a two-stream wavelet enhanced U-net (TSWEU) model to learn the connection between the stripe and restored image. Zhong et al. [51] proposed a satellite-ground integrated destriping network (SGIDN), which considers both the spectral and spatial intrinsic features. It alleviates data dependence by an integrated satellite-ground strategy and possesses a simple CNN structure incorporating different convolutional layers. In addition, an unsupervised method combined with the adversarial learning strategy was presented to remove stripe in the hyperspectral image [52]. A multiscaled column-spatial correction network (CSCNet) was introduced by Li et al. [10] to remove the stripe noise, which restores images from both local and global perspectives at multiple scales, including structural and contextual image information. In addition, CSCNet designs the fusion module to effectively select the significant signals.

III. Methodology

A. Overview

To remove diverse stripe noise in multispectral remote sensing images and enhance the uniformity of the entire image, we propose a novel destriping method, the PRNet, which gradually restores the clean image in an iterative manner. The structure of the proposed PRNet is shown in Fig. 1. It consists of two major components: PSM and MRM. To exploit the local spatial correlation, the PSM is designed to explore spatial information within and between local regions. To effectively use the structural information of the original image, the local spatial correlation features, and the clean background information, the MRM is designed to extract features within local spatial regions using a carefully designed spatial attention module. The MRM implements stripe removal in multiple timesteps, capturing the relevance of detailed image information among the processed results of adjacent timesteps.

Specifically, the stripe removal result generated by the previous timestep is fed to the PSM to extract local spatial correlation features at three scales. The PSM splits the original degraded image into multiscale patch sequences along the column direction, leveraging the unique structural characteristic of stripe noise. The outputs of PSM, original degraded image, and stripe removal results generated by the previous timestep are then concatenated and passed through the MRM for a second round of destriping. During the iterative destriping process, the original degraded image is adopted as the input for the first iteration instead of the stripe removal results. The stripe removal results that are inputs to both PSM and MRM are updated before each iteration timestep. This process is repeated several times until a clean image is obtained. To optimize the parameters of the model, structural similarity (SSIM) is used in PRNet.

B. Patch-Based Sequence Module

To capture the local spatial correlation and enhance the uniformity of the entire image, we propose the PSM in the lower left part of Fig. 1. The PSM consists of three patch-based sequence networks (PSNs) to process different patch sequences. Each PSN uses the ConvLSTM to capture the local spatial correlation between patches in sequences, and its details are presented in Section III-D1.

To effectively capture local spatial correlations with different scales, we split the input image from coarse to fine,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

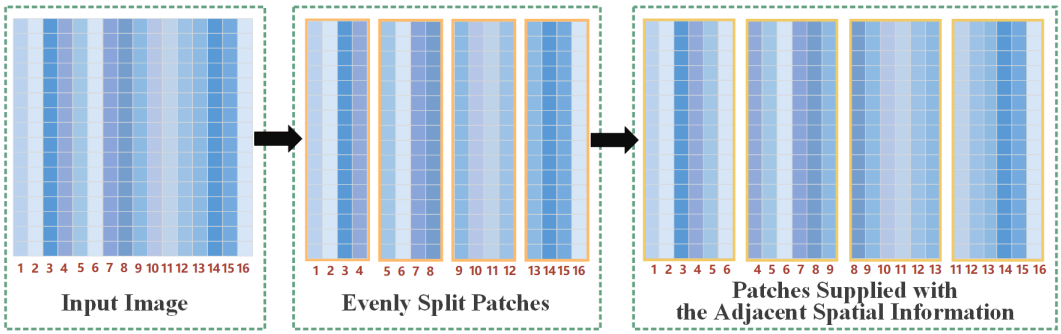


Fig. 2. Generation of the patch sequence. First, the input image is evenly split into several patches along with the column direction. Then, each patch is supplied with the adjacent spatial information to further strengthen the local spatial correlation.

generating patch sequences $S_k(k = 1, 2, 3)$ of different sizes. Specifically, the input feature $\mathbf{C} \in \mathbb{R}^{H\times W\times 1}$ generated from the previous timestep is evenly split into patch sequences S_k as follows:

$$\mathbf{C} \Longrightarrow S_k(k = 1, 2, 3) \tag{2}$$

where $\Longrightarrow$ is the split operation along the column direction, and $S_k \in \mathbb{R}^{H\times W/2^k\times 1}$ are the obtained patch sequences. To enhance the correlation among patches, as shown in Fig. 2, we supplement each patch with neighboring regions $\mathbf{m} \in \mathbb{R}^{H\times(W/(4\times k))\times 1}$ selected along the left and right directions from the original image. For patches at the beginning and end of the sequence, the regions $\mathbf{m} \in \mathbb{R}^{H\times(W/(2\times k))\times 1}$ are selected along the right or left direction on the original image. This generates patch sequences $S_k \in \mathbb{R}^{H\times((3\times W)/(2\times k))\times 1}$. The supplement of spatial information results in overlap between adjacent patches, which further strengthens their local spatial correlation.

After the patch sequences are generated, they are fed into three PSNs, respectively. In each PSN, a convolutional layer followed by a ReLU activation is placed before and after the ConvLSTM to refine features and adjust the number of channels. The parameters of the former convolutional layer are (1, 32, 3, 1, 1), representing the size of the input channel, output channel, convolutional kernel, stride, and padding, respectively. The parameters of the latter are (32, 1, 3, 1, 1). The yielded patch sequences $S_k' \in \mathbb{R}^{H\times((3\times W)/(2\times k))\times 1}$ are fused to generate the output of PSM, noted as f_k. Due to the overlap between adjacent patches, the patch sequences S_k' resulting from the three ConvLSTMs cannot be concatenated directly. The output features f_k are obtained by fusing the overlapped parts. Specifically, we no longer change the parts that do not overlap, while calculating the mean value of the overlapped parts as the final features.

C. Main Recurrent Module

In PRNet, a progressive destriping strategy is presented to handle different degrees of stripe noise, in which the MRM is responsible for gradually restoring the clean image. As shown in the lower right part of Fig. 1, MRM is designed with two parts, namely, the ConvLSTM and spatial attention block. Since the stripe noise is removed gradually during multiple iterations, the stripe removal results generated by multiple iteration timesteps can be viewed as a sequence with small changes in detailed information between images. To effectively use the correlation of detailed image information, similar to PSM, ConvLSTM is leveraged to implement the progressive destriping strategy given the powerful capacity of processing contextual information in a sequence. Moreover, we introduce a spatial attention module focusing on the spatial information within the local regions, which is complementary to PSM that explores the correlation between local regions and further improves the destriping ability of the model.

PRNet uses a progressive destriping strategy to handle different degrees of stripe noise, in which the MRM is responsible for gradually restoring the clean image. As shown in the lower right part of Fig. 1, MRM is composed of two parts: the ConvLSTM and spatial attention block. Since the stripe noise is removed gradually during multiple iterations, the stripe removal results generated by multiple iteration timesteps can be viewed as a sequence with small changes in detailed information between images. To effectively use the correlation of detailed image information, ConvLSTM is used to implement the progressive destriping strategy, given its powerful capacity of processing contextual information in a sequence. Moreover, PRNet introduces a spatial attention module that focuses on the spatial information within local regions. This module is complementary to PSM, which explores the correlation between local regions and further improves the destriping ability of the model.

To implement MRM, we first concatenate the original degraded image $\mathbf{I} \in \mathbb{R}^{H\times W\times 1}$, the features $f_k \in \mathbb{R}^{H\times W\times 1}$ $(k = 1, 2, 3)$, and the stripe removal result $\mathbf{C} \in \mathbb{R}^{H\times W\times 1}$ to form the input feature $f_{\mathrm{MI}} \in \mathbb{R}^{H\times W\times 5}$ for MRM. We then feed f_{MI} to a convolutional layer (5, 32, 3, 1, 1) followed by a ReLU activation, which is used to refine the feature and change channels. The yielded features are then sent to the ConvLSTM. The output features of ConvLSTM pass through one spatial attention block to exploit the spatial information within local regions. Finally, similar to the start of MRM, we use a convolutional layer (32, 1, 3, 1, 1) followed by a ReLU activation to refine the feature and change channels. We then add the original degraded image to obtain the stripe removal image of the single timestep $\mathbf{C} \in \mathbb{R}^{H\times W\times 1}$. The entire process can be written as follows:

$$\mathbf{C} = \mathrm{conv1}\times 1(\mathrm{SAB}(\mathrm{ConvLSTM}(\mathrm{conv1}\times 1(f_{\mathrm{MI}})))) \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

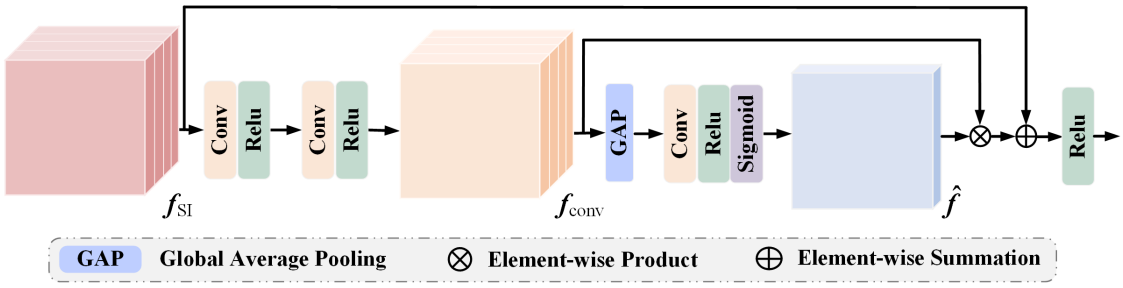


Fig. 3. Structure of the spatial attention block.

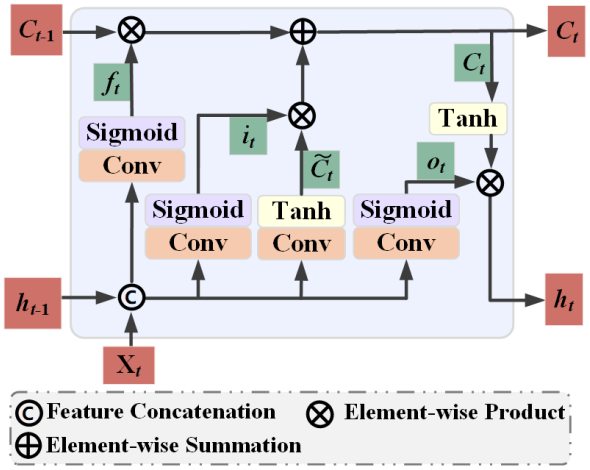


Fig. 4. Structure of ConvLSTM.

where $\text{conv}_{1\times1}(\cdot)$ is the 1×1 convolutional layer, and ConvLSTM$(\cdot)$ and SAB$(\cdot)$ stand for ConvLSTM and spatial attention block, respectively.

As for the spatial attention block, its structure is as shown in Fig. 3. The input feature $f_{SI} \in \mathbb{R}^{H\times W\times 32}$ first passes through two convolutional layers (32, 32, 3, 1, 1) followed by a ReLU activation. Subsequently, the global average pooling (GAP) operation is implemented to compress the yielded feature $f_{conv} \in \mathbb{R}^{H\times W\times 32}$ in channel dimension. Then, we obtain the spatial correction coefficients $\hat{f} \in \mathbb{R}^{H\times W\times 1}$ by adopting one convolutional layer (1, 1, 5, 1, 2) followed by a ReLU activation to process the generated feature. The process can be written as

$$f_{conv} = \text{conv}_{3\times3}(\text{conv}_{3\times3}(f_{SI})) \tag{4}$$

$$\hat{f} = \text{conv}_{5\times5}(\text{GAP}_s(f_{conv})). \tag{5}$$

Here, $\text{conv}_{3\times3}(\cdot)$ and $\text{conv}_{5\times5}(\cdot)$ are the 3×3 and 5×5 convolutional layers, respectively. $\text{GAP}_s(\cdot)$ is the spatial global average pooling. By designing a larger convolutional kernel size, the receptive field is extended to enable the model to capture more spatial information. To enhance the coefficient map $\hat{f}$, the sigmoid gating is used to refine the generated features. Finally, the uniformity of features is strengthened by multiplying f_{conv} with the coefficients $\hat{f}$. The output feature of the spatial attention block is generated by adding the input feature f_{SI} according to the residual structure design. We adopt

Algorithm 1 Training process

input : The striped image **I** (size of $H \times W \times 1$),
The clean image **I*** (size of $H \times W \times 1$),
Epoch (m), Iteration (n);
output: The trained model

for $i \leftarrow 1$ **to** m **do**
  for $j \leftarrow 1$ **to** n **do**
    if $j == 1$ **then**
      Stripe removal result **C** $\leftarrow$ Image **I**
    end
    S$_1$ = [p$_{11}$, p_{12}], **S**$_2$ = [p$_{21}$, p_{22}, p_{23}, p_{24}], **S**$_3$ = [p$_{31}$, p_{32}, ..., p_{38}] $\leftarrow$ Segment the Stripe removal result **C**
    for $b \leftarrow 1$ **to** 2 **do**
      S$_1'$[b] $\leftarrow$ input **S**$_1$[b] to **PSN1**
    end
    $f_1 \leftarrow$ fuse **S**$_1'$
    for $c \leftarrow 1$ **to** 4 **do**
      S$_2'$[c] $\leftarrow$ input **S**$_2$[c] to **PSN2**
    end
    $f_2 \leftarrow$ fuse **S**$_2'$
    for $d \leftarrow 1$ **to** 8 **do**
      S$_3'$[d] $\leftarrow$ input **S**$_3$[d] to **PSN3**
    end
    $f_3 \leftarrow$ fuse **S**$_3'$
    Stripe removal result **C** $\leftarrow$ $Input${**I**, **C**, f$_1$, f$_2$, f$_3$} to **MRM** (Main Recurrent Module).
  end
  Calculate the error between **C** and **I*** according to the loss function.
  Update the network parameters.
end

$\hat{f}$ to reflect the overall tone at the feature level, as follows:

$$f_{out} = \hat{f} \otimes f_{conv} \oplus f_{SI} \tag{6}$$

where $\otimes(\cdot)$ and $\oplus(\cdot)$ are elementwise summation and multiplication. $f_{out} \in \mathbb{R}^{H\times W\times 1}$ is the output of the spatial attention block.

D. Implementation Details

1) Convolutional LSTM: As a particular recurrent neural network (RNN) structure, LSTM has been shown to handle the sequential correlation stably, relying on the gate mechanism

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

to control the flow and loss of features. The original LSTM is suitable for the 1-D vector, while the spatial information is not taken into consideration. Subsequently, ConvLSTM was proposed to address the above drawback, which is an extension of the original LSTM and uses convolutional operations in all the state conversions. The key equations of ConvLSTM are as follows:

$$f_t = \sigma\left(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f\right) \tag{7}$$
$$i_t = \sigma\left(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i\right) \tag{8}$$
$$\widetilde{C}_t = \tanh\left(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c\right) \tag{9}$$
$$C_t = f_t \otimes C_{t-1} + i_t \otimes \widetilde{C}_t \tag{10}$$
$$o_t = \sigma\left(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o\right) \tag{11}$$
$$h_t = o_t \otimes \tanh(C_t) \tag{12}$$

where σ is the Sigmoid activation, $*$ refers to the convolutional operator, and $\otimes$ indicates the elementwise product. There are 3-D tensors in ConvLSTM, including all the inputs $X_1, X_2, \ldots, X_t$, cell outputs $C_1, C_2, \ldots, C_t$, hidden states $H_1, H_2, \ldots, H_t$, and gates i_t, f_t, o_t.

In PRNet, ConvLSTM is used to explore the correlation between elements in a sequence, namely, the detailed image information correlation in MRM and the local spatial correlation in PSM. The structure of ConvLSTM used in this study is shown in Fig. 4. It is designed according to (7), and the parameters of the convolutional layer are set to (64, 32, 3, 1, 1).

2) Training Details: The detailed pseudocode is shown in Algorithm 1. In PRNet, the number of PSN is set to 3, and the iteration number of MRM is set to 6. The related ablation analyses are given in Section IV-F. In this study, the University of Houston dataset is used to generate the training samples. Specifically, we selected 10000 image patches with the size of 128 × 128, and these samples were then normalized to [0, 1]. The epoch and batch size were set to 100 and 24, respectively. We initially applied a learning rate of 0.001, which was reduced to 0.2 of the current learning rate over 30, 50, and 80 training epochs. In addition, the ADAM optimizer was used to train the proposed model.

IV. Experiments

In this section, we compared the proposed PRNet against ten classic destriping methods, including traditional methods MM [17], ASSTV [14], LR-based single-image decomposition (LRSID) model [36], TV regularization and group sparsity model (TVGS) [53], proximal alternating direction method of multipliers (PADMM) [54], adaptive material matching (MAM) [55], and deep-learning-based methods ICSRN [11], SSGN [12], TSWEU [33], and CSCNet [10] on multiple datasets. The nondata-driven approaches, including MM, ASSTV, LRSID, TVGS, PADMM, and MAM, were experimented on a personal laptop with the following configuration, i.e., the MATLAB 2016a platform with a 1.6-Hz Intel i5-10210U CPU and with 8-GB RAM. Data-driven methods, including ICSRN, SSGN, CSCNet, and PRNet, were trained and tested on the workstation with PyTorch 1.1.0 framework and a 3.8-GHz Intel i1-8700k CPU and with 128-GB RAM, and an NVIDIA GTX1080Ti GPU.

Moreover, to further verify the effectiveness of PRNet, the performance of the models was quantitatively analyzed. The quantitative evaluation includes the signal-to-noise ratio (SNR), peak SNR (PSNR), SSIM, spectral angular distance (SAD), and mean relative deviation (MRD). We also investigated the nonuniformity of the restored images. In addition, the mean cross-profile analysis of destriping results was provided to assess the effects of different methods.

A. Simulated Image Destriping

1) Simulated Data Preparation: Three remote sensing images were adopted as clean images to generate simulated datasets, namely, HYDICE Urban, Pavia University, and Washington dc Mall. They are noted as Urban, PaviaU, and dc for convenient references in Seciton IV. By referring to [33], three ways of adding simulated stripes were investigated, including the multiplicative, additive, and mixed stripe noise, which were added on Urban, PaviaU, and dc images, respectively. Their generation process can be expressed by the following formulas:

$$V'_{l,j}(\varphi) = A_{l,j} \times v_{l,j}(\varphi), \tag{13}$$
$$V'_{l,j}(\varphi) = v_{l,j}(\varphi) + C_{l,j}, \tag{14}$$
$$V'_{l,j}(\varphi) = A_{l,j} \times v_{l,j}(\varphi) + C_{l,j} \tag{15}$$

where $v_{l,j}(\varphi)$ represents the pixels that are on row l and column j. To model the directional property of stripe noise, $A_{l,j}$ and $C_{l,j}$ are the multiplicative and additive noise values, which are random values. We allocated the same values of $A_{l,j}$ and $C_{l,j}$ for the same column. In this study, the range of $A_{l,j}$ was set to [0.5, 1.2] for multiplicative noise [see (13)]. The range of $C_{l,j}$ was set to [−650, −600] and [600, 650] for additive noise [see (14)]. The ranges of $A_{l,j}$ and $C_{l,j}$ were set to [0.1, 1.5] and [0, 600] for mixed noise [see (15)]. Figs. 5(a)–7(a) display the degraded images. To make a fair comparison, we used the same training data and protocols to retrain the data-driven models, and the original implementations of these models were referred throughout the whole training and testing phases.

2) Qualitative Evaluation: On the simulated images, the visualized results are shown in Figs. 5(b)–(l)–7(b)–(l), indicating that PRNet achieves satisfactory effects compared with other algorithms. Specifically, the destriping performance of MM is inferior because of the considerable difference in ground objects between columns. While removing some of the stripe noise, it also results in changing the original image information. In Figs. 5(b)–7(b), we can observe that MM changes the brightness of some areas after correcting the image. This is caused by the assumption that the statistics in each column are always equal, which does not conform to the real situation. As for MAM, the relatively satisfactory destriping performance can be obtained when facing the scenarios with the simple material distribution, as shown in Figs. 5(g) and 6(g). We can see from Fig. 7(g) that the complicated distribution of the ground target still leads to the failure cases in the restored results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

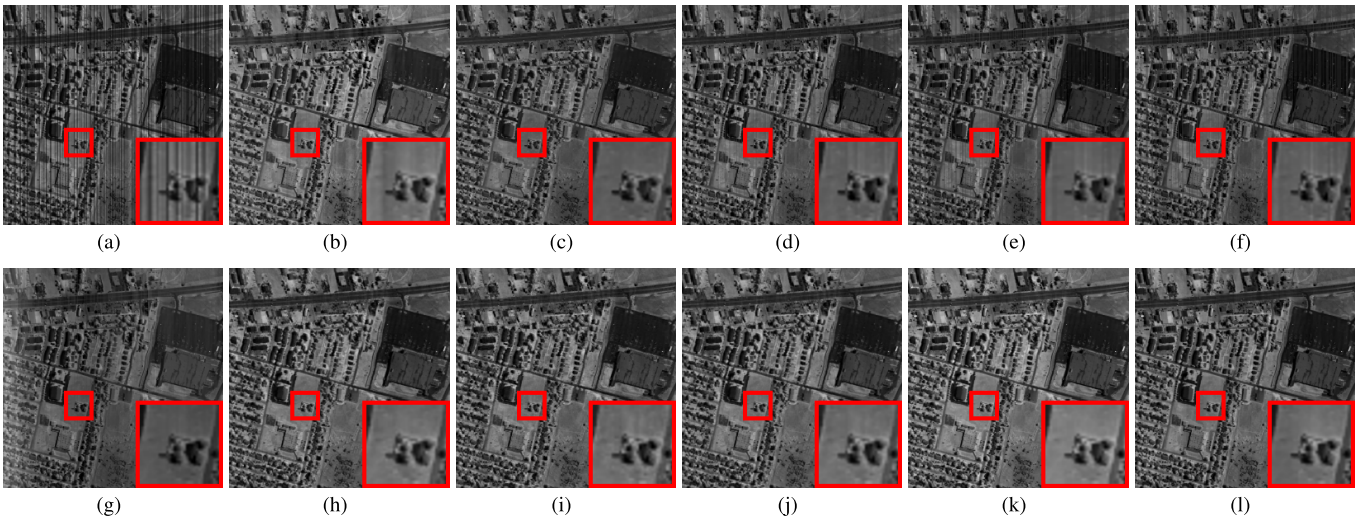


Fig. 5. Corrected urban images for band 93. (a) Degraded image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

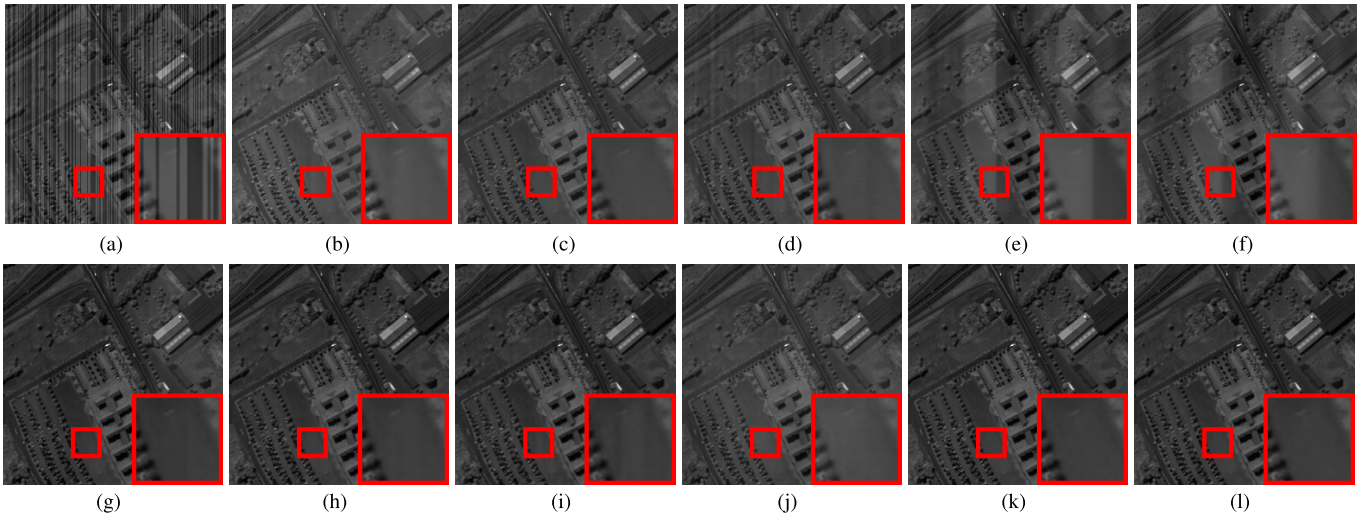


Fig. 6. Corrected PaviaU images for band 75. (a) Degraded image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

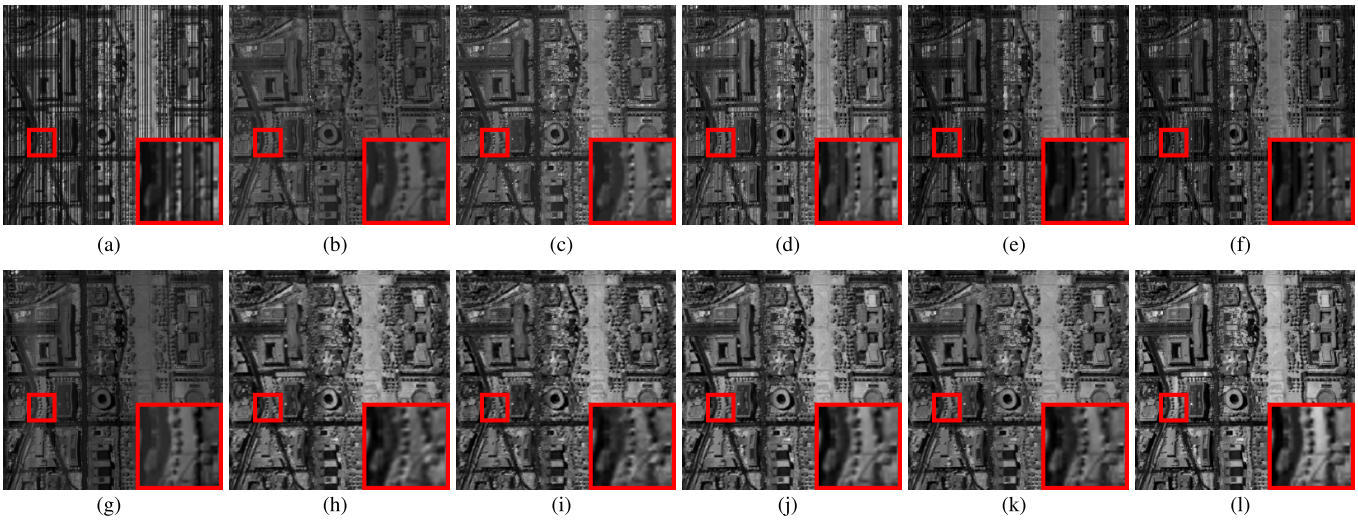


Fig. 7. Corrected dc images for band 80. (a) Degraded image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

As for optimization-based methods, from Figs. 5(c)–(f)–7(c)–(f), we can observe that the destriping performance of ASSTV is somewhat satisfactory in view of the adoption of the TV model to eliminate the intercolumn differences resulted from the stripe noise. However, LRSID fails to perform stripe noise removal in terms of additive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE ASSESSMENTS OF DIFFERENT METHODS ON THREE SIMULATED DATASETS

Image	Index	Striped	MM	ASSTV	LRSID	TVGS	PADMM	MAM	ICSRN	SSGN	TSWEU	CSCNet	PRNet
Urban	MSNR (↑)	9.7444	10.3065	12.1131	12.3988	11.4838	11.4488	11.6209	16.7828	15.5553	16.2438	16.5862	17.7001
	MPSNR (↑)	24.3855	25.4855	27.0281	27.2352	26.3257	26.2603	26.6761	29.7844	28.2180	28.4498	28.6603	30.3965
	MSSIM (↑)	0.7600	0.9227	0.9332	0.9267	0.9005	0.9011	0.9157	0.9452	0.9256	0.9475	0.9571	0.9655
	MSAD (↓)	0.2550	0.1434	0.1252	0.1367	0.1768	0.1875	0.1596	0.1377	0.1556	0.1304	0.1121	0.1282
PaviaU	MSNR (↑)	9.2015	11.0652	14.0487	13.7447	12.0596	12.0639	17.1692	24.7123	22.1203	30.8570	31.0926	31.3422
	MPSNR (↑)	21.4048	24.1418	26.6219	26.3143	24.6455	24.6712	30.4649	38.8370	36.2160	44.5682	45.3926	45.4119
	MSSIM (↑)	0.4359	0.8675	0.9129	0.9120	0.8970	0.9119	0.9675	0.9866	0.9747	0.9937	0.9953	0.9969
	MSAD (↓)	0.3418	0.2499	0.1712	0.1491	0.2270	0.2344	0.1278	0.0756	0.0889	0.0672	0.0635	0.0623
DC	MSNR (↑)	5.4576	4.7979	5.4834	5.8774	6.2739	6.2360	5.5836	8.7067	8.8080	8.9773	8.0117	9.8228
	MPSNR (↑)	18.2874	18.9920	18.4272	18.8886	19.5734	19.5572	18.6445	20.2071	20.1208	20.3787	19.1728	21.1465
	MSSIM (↑)	0.4838	0.6758	0.6668	0.6729	0.6640	0.6647	0.6867	0.6839	0.6798	0.7213	0.7111	0.7483
	MSAD (↓)	0.5484	0.5277	0.5150	0.5144	0.5172	0.5242	0.5160	0.4874	0.4885	0.5062	0.5342	0.4886

LR constraints, remaining the residual stripe noise in Figs. 5(d)–7(d). For TVGS and PADMM, their destriping results are less satisfactory. As shown in Fig. 5(e) and (f), the stripe noise is eliminated, while the nonuniformity phenomenon appears in the restored images. In addition, they cannot handle the severe stripe noise, and there are a large number of residual stripes after destriping, which is significant in Fig. 7(e) and (f).

Compared with conventional methods, from Figs. 5(h)–(l)–7(h)–(l), the deep-learning-based methods obtained advanced performance. However, as shown in Figs. 5(h) and (i) and 6(h) and (i), ICSRN and SSGN are unable to completely remove the severe stripes. For the mixed stripe noise, they destroyed the structural feature of the degraded image, resulting in the fuzzy phenomenon in the restored image, which can be observed in Fig. 7(h) and (i). From the visualized destriping results of the simulated data, PRNet can successfully remove stripe noise while preserving image detail information through the carefully designed destriping strategy.

3) Quantitative Evaluation: To quantify the destriping effect of ten approaches, we used the mean value of SNR, PSNR, SSIM, and SAD for the whole dataset to measure the corrected images, which were noted as MSNR, MPSNR, MSSIM, and MSAD, respectively. The definitions of adopted evaluation indicators are as follows:

$$\mathbf{SNR} = 10\log 10\left(\frac{\sum_{x=0}^{m-1}\sum_{y=0}^{n-1}(x(i,j))^2}{\sum_{x=0}^{m-1}\sum_{y=0}^{n-1}(x(i,j)-y(i,j))^2}\right) \tag{16}$$

$$\mathbf{MSE} = \frac{1}{mn}\left(\sum_{x=0}^{m-1}\sum_{y=0}^{n-1}(x(i,j)-y(i,j))^2\right) \tag{17}$$

$$\mathbf{PSNR} = 10\log 10\left(\frac{\mathrm{MAX}_I^2}{\mathbf{MSE}}\right)$$

$$\mathbf{SSIM} = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + +c_1)(\sigma_x^2 + \sigma_y^2 + +c_2)} \tag{18}$$

$$\mathbf{SAD} = \cos^{-1}\frac{xy}{\|x\|\|y\|} = \cos^{-1}\frac{x_1y_1 + x_2y_2 + \cdots + x_ny_n}{\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}\times\sqrt{y_1^2 + y_2^2 + \cdots + y_n^2}}. \tag{19}$$

Here, $x(i,j)$ and $y(i,j)$ represent the pixel values with coordinate (i,j) in the clean image and the restored image, respectively. m and n refer to the size of an image, namely, the number of rows and columns. μ_x, μ_y, σ_x, and σ_y denote the mean and variance of images x and y, respectively. σ_{xy} is the covariance of images x and y. $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ represent the pixel values of two images on different spectral bands. The first two indicators are often adopted to evaluate the image quality after denoising, SSIM measures the loss of structural information, and SAD reflects the degree of spectral distortion. The corresponding quantitative assessments of three simulated images are shown in Table I. In addition, to verify the validity of the method affected by different noise intensities, we compared four intensities of additive noise for the PaviaU dataset, and the quantitative assessments are listed in Table II.

As indicated in Table I, PRNet achieves the best MSNR, MPSNR, and MSSIM on three simulated datasets, indicating that our method provides the restored images with higher quality and clearer structural information of the original image. As for MSAD, the proposed PRNet has the lowest values on the PaviaU dataset and a relatively small value on the Urban and dc datasets, which further demonstrates that spectral distortion is effectively constrained. From Table I, we can observe that the deep-learning-based method has the greatest advantage over other methods in removing additive noise, resulting from the additive noise that does not contain image information, which is more conducive to the deep-learning-based methods to separate out the noise. The multiplicative coefficients in multiplicative and mixed noise make the added noise contain image information, which further complicates the difficulty of separating the noise. From Table II, we can see that PRNet possesses the optimal performance in four cases, which confirms the robustness of our model to handle different intensities of stripe noise. From the perspective of four

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE ASSESSMENTS OF THE PAVIAU IMAGE WITH DIFFERENT STRIPE NOISE INTENSITIES

Intensity	Index	MM	ASSTV	LRSID	TVGS	PADMM	MAM	ICSRN	SSGN	TSWEU	CSCNet	PRNet
(-50,50)	MSNR (↑)	13.0033	25.5134	32.2511	30.6266	33.4494	16.8712	31.8039	25.6457	33.5194	34.7197	36.0623
	MPSNR (↑)	27.8684	39.5322	46.2925	44.6825	47.4844	30.2365	45.9193	39.7924	47.4621	48.8347	50.1804
	MSSIM (↑)	0.9391	0.9883	0.9940	0.9916	0.9977	0.9707	0.9955	0.9834	0.9968	0.9979	0.9984
	MSAD (↓)	0.1010	0.0929	0.0518	0.0315	0.0254	0.1400	0.0278	0.0458	0.0257	0.0250	0.0200
(-100,-50) (50,100)	MSNR (↑)	13.1915	23.5900	28.0371	25.6425	26.0671	16.6241	27.8437	24.0791	32.8994	34.0696	36.0204
	MPSNR (↑)	27.8039	37.4781	41.9399	39.5653	39.9708	29.9177	41.9406	38.2302	46.4934	48.2238	50.1385
	MSSIM (↑)	0.9365	0.9844	0.9922	0.9859	0.9925	0.9675	0.9905	0.9776	0.9956	0.9981	0.9987
	MSAD (↓)	0.1074	0.1014	0.0597	0.0566	0.0557	0.1235	0.0469	0.0613	0.0375	0.0308	0.0284
(-200,-100) (100,200)	MSNR (↑)	13.0893	20.7009	23.3474	20.7661	20.6486	16.3078	25.3968	21.9307	30.0948	31.3989	32.8421
	MPSNR (↑)	27.2284	34.3792	37.0339	34.4765	34.3480	29.5892	39.5960	35.9513	44.4786	45.5268	46.9997
	MSSIM (↑)	0.9243	0.9749	0.9835	0.9731	0.9806	0.9651	0.9873	0.9682	0.9962	0.9968	0.9979
	MSAD (↓)	0.1281	0.1194	0.0801	0.0977	0.1013	0.1466	0.0631	0.0841	0.0513	0.0488	0.0431
(-300,-200) (200,300)	MSNR (↑)	12.4898	18.7971	19.6398	17.2929	17.1278	16.4591	24.3164	21.3735	29.0138	30.9732	32.6559
	MPSNR (↑)	26.1181	32.1792	33.0335	30.7096	30.5383	29.7071	38.2622	35.3980	43.8927	45.0378	46.7871
	MSSIM (↑)	0.9074	0.9628	0.9708	0.9568	0.9656	0.9637	0.9842	0.9638	0.9953	0.9970	0.9979
	MSAD (↓)	0.1380	0.1259	0.0904	0.1304	0.1377	0.1487	0.0654	0.0862	0.0574	0.0432	0.0409

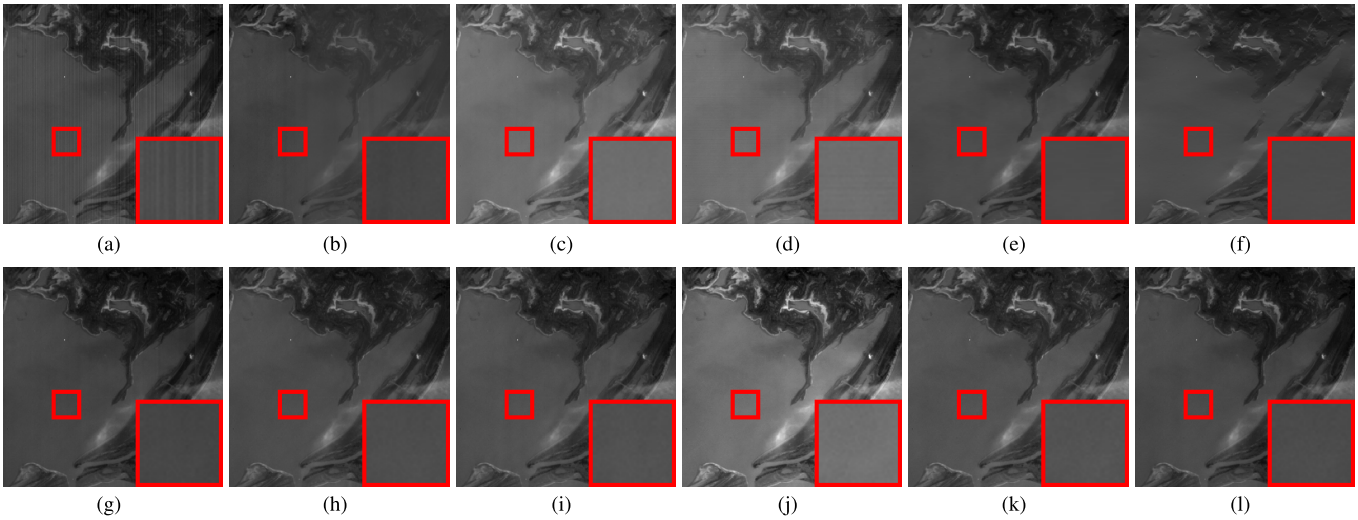


Fig. 8. Corrected CHRIS_AM images for band 4. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

indicators, the destriping capabilities of most methods deteriorate as the intensity of stripe noise increases, in which deep learning methods have better generalization ability compared with the traditional ones. As listed in Table II, it can be seen that the destriping performance of MM and MAM is stable. This is because they rely on the statistics of each column and the pixels of the same material between columns to remove stripe noise, while the statistics and same material pixels are not impacted by the noise intensity. Overall, the proposed PRNet significantly improves the image quality after removing stripe noise with different types and intensities.

B. Real Image Destriping

1) Real Data: To observe the applicability of different methods, the destriping results on three real data were further investigated. Among them, two are available CHRIS_AM and CHRIS_UK images. The last one was generated by the full spectrum airborne hyperspectral imager (FAHI) [56]. The shortwave-infrared (SWIR) image generated by FAHI is used to test the destriping performance of ten methods. The original striped images are shown in Figs. 8(a)–11(a).

2) Qualitative Evaluation: The visualized results on real images are shown in Figs. 8(b)–(k)–11(b)–(k). It can observe that the proposed PRNet provides advanced destriping performance on real images with more complicated stripe noise distribution compared with other methods. Fig. 11(b) is similar to the performance shown on the simulated data; MM is more suitable for the distributions of ground objects that are consistent between columns. However, from Figs. 8(b)–10(b), we can observe that the original image information is damaged after the correction. As shown in Figs. 8(g)–10(g), MAM generated satisfactory results, while it also caused the nonuniformity of the local region in Fig. 9(g). This issue is because the material matching pixels cannot be selected correctly when dealing with the complex distribution of material.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

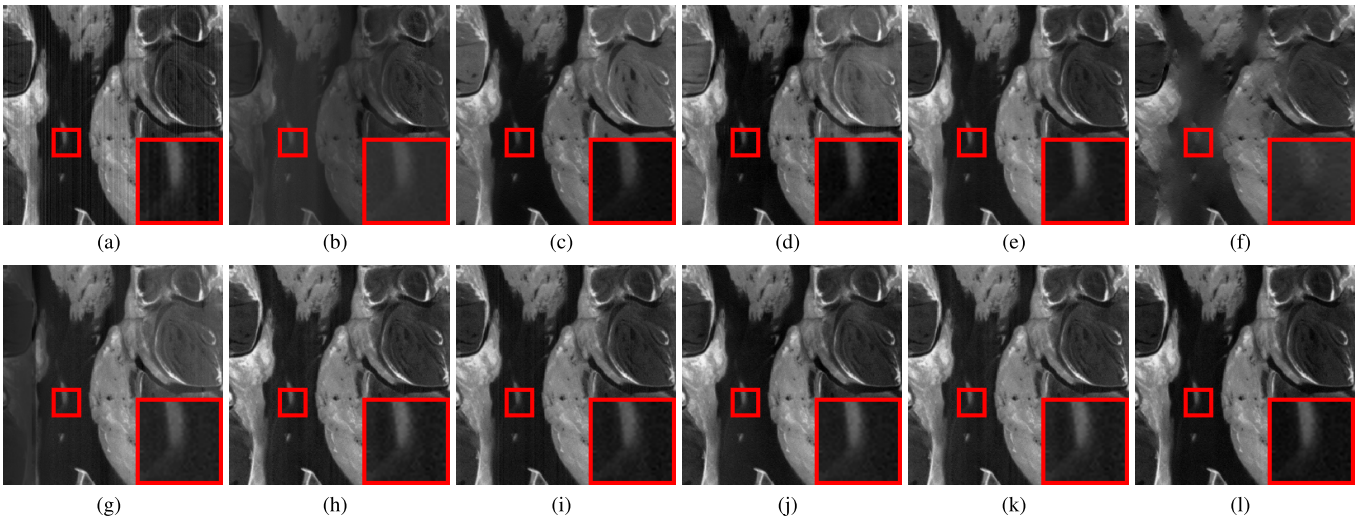


Fig. 9. Corrected CHRIS_UK images for band 1. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

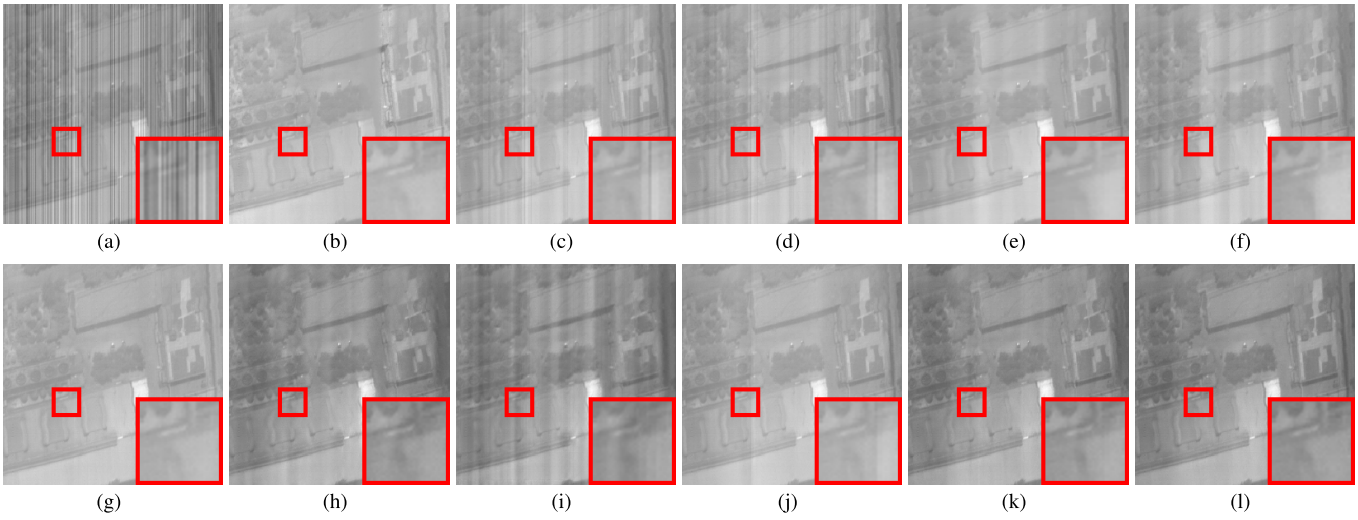


Fig. 10. Corrected SWIR images for band 85. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

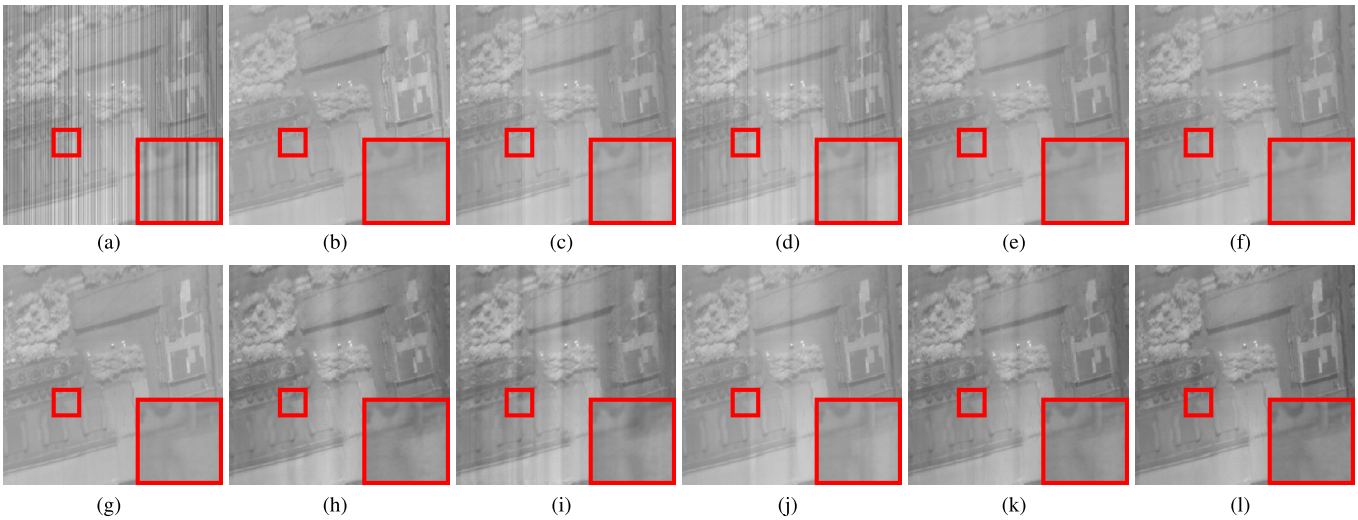


Fig. 11. Corrected SWIR images for band 504. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

From Figs. 8(c)–(f)–11(c)–(f), the optimization-based methods have a decent stripe removal outcome for slight stripe noise. In Fig. 9(d) and (f), the quality of restored images corrected by LRSID and PADMM is degraded, including the horizontal texture feature and fuzzy phenomenon. As for images disturbed by the severe noise, most of them

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MRD OF DIFFERENT METHODS ON REAL DATASETS

Image	MM	ASSTV	LRSID	TVGS	PADMM	MAM	ICSRN	SSGN	TSWEU	CSCNet	PRNet
CHRIS_AM	0.0211	0.0880	0.0811	0.0101	0.0235	0.0244	0.0706	0.0569	0.0417	0.0274	0.0157
CHRIS_UK	0.0525	0.0444	0.0521	0.0201	0.0664	0.0834	0.1178	0.0894	0.0471	0.0332	0.0278
SWIR (band 85)	0.0113	0.0832	0.0828	0.0922	0.2176	0.1177	0.1017	0.0935	0.0927	0.1057	0.1070
SWIR (band 504)	0.1045	0.0742	0.0715	0.1509	0.1874	0.3032	0.1783	0.1701	0.1873	0.1936	0.1934

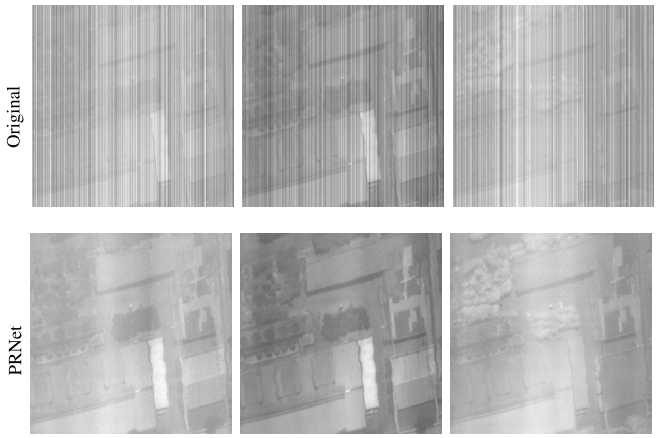


Fig. 12. Destriping results of PRNet on SWIR images. (Left to right) Bands 37, 79, and 508.

just attenuate the stripe noise to some extent, rather than completely remove them, especially in Fig. 10(c)–(f) and Fig. 11(c), (d), and (f).

As for deep-learning-based methods, they achieve promising destriping performance in comparison to other approaches, in Figs. 8(h)–(i)–11(h)–(i). However, from Figs. 10(h) and (i) and 11(h) and (i), the restored images generated by ICSRN and SSGN are less satisfactory when facing complicated and severe stripe noise. As shown in Figs. 10(j) and (k) and 11(j) and (k), CSCNet and TSWEU can further remove the stripe noise compared with ICSRN and SSGN, while their destriping effect is inferior to PRNet.

In addition, Fig. 12 shows the destriping results of SWIR images for bands 37, 79, and 508. It can be observed that the image information is heavily disturbed by the noise, and the targets are almost drowned out by the stripe interference in the visualization. Nevertheless, the proposed method is able to recover the image information and greatly improve the image uniformity, demonstrating the effectiveness of PRNet on noise-heavy data. In general, the proposed PRNet effectively restores the degraded images disturbed by the stripe noise with different intensities in practice, which is more complicated compared with the simulated data, relying on the carefully designed MRM and PSM.

3) Quantitative Evaluation: The information of degraded images should be retained to the greatest extent after destriping. Therefore, we used MRD to quantify the similarity between the restored and degraded images, which is defined as [21]

$$\mathrm{MRD} = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\frac{|y(i,j)-x(i,j)|}{x(i,j)}. \tag{20}$$

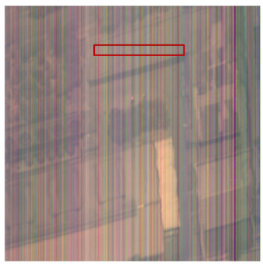


Fig. 13. Ground target region in SWIR image.

Here, $x(i,j)$ and $y(i,j)$ denote the pixel values with coordinate (i,j) in the degraded image and the restored image, respectively. m and n refer to the size of an image, namely, the number of rows and columns. A lower MRD value implies a higher similarity between the restored images and degraded images. It is worth noting that a low MRD value is only meaningful after the stripe noise has been successfully removed.

The calculation results are presented in Table III for the ten methods. It can be noted that the MRD of PRNet is only slightly higher than TVGS on CHRIS_AM for band 4 and CHRIS_UK for band 1, which indicates that the distortion of the original information resulting from PRNet is small during image correction. For bands 85 and 504 of the SWIR image, only PRNet and MAM can remove the stripe noise completely. From III, we can see that PRNet has a low MRD compared with MAM on these bands, which further demonstrates that PRNet can effectively perform the removal of the complex stripe noise while retaining the image information.

C. Image Uniformity

The uniformity of the degraded images is supposed to be promoted by destriping. Consequently, the destriping effect of approaches can be verified quantitatively by calculating the nonuniformity of the restored images as follows:

$$U = \sqrt{\frac{\sum_{i=1}^{N_s}\left(S_i-\bar{S}\right)^2}{N_s}}. \tag{21}$$

Here, U stands for the image nonuniformity, N_s is the number of the whole samples, S_i is the value of pixel i, and $\bar{S}$ refers to the mean value of S_i. The small nonuniformity values suggest that the uniformity of the restored image is significantly improved. We selected the uniform region with the size of 10×90 from SWIR images highlighted by the red box in Fig. 13.

The computational values of all the methods are presented in Table IV. It can be observed that PRNet has the smallest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

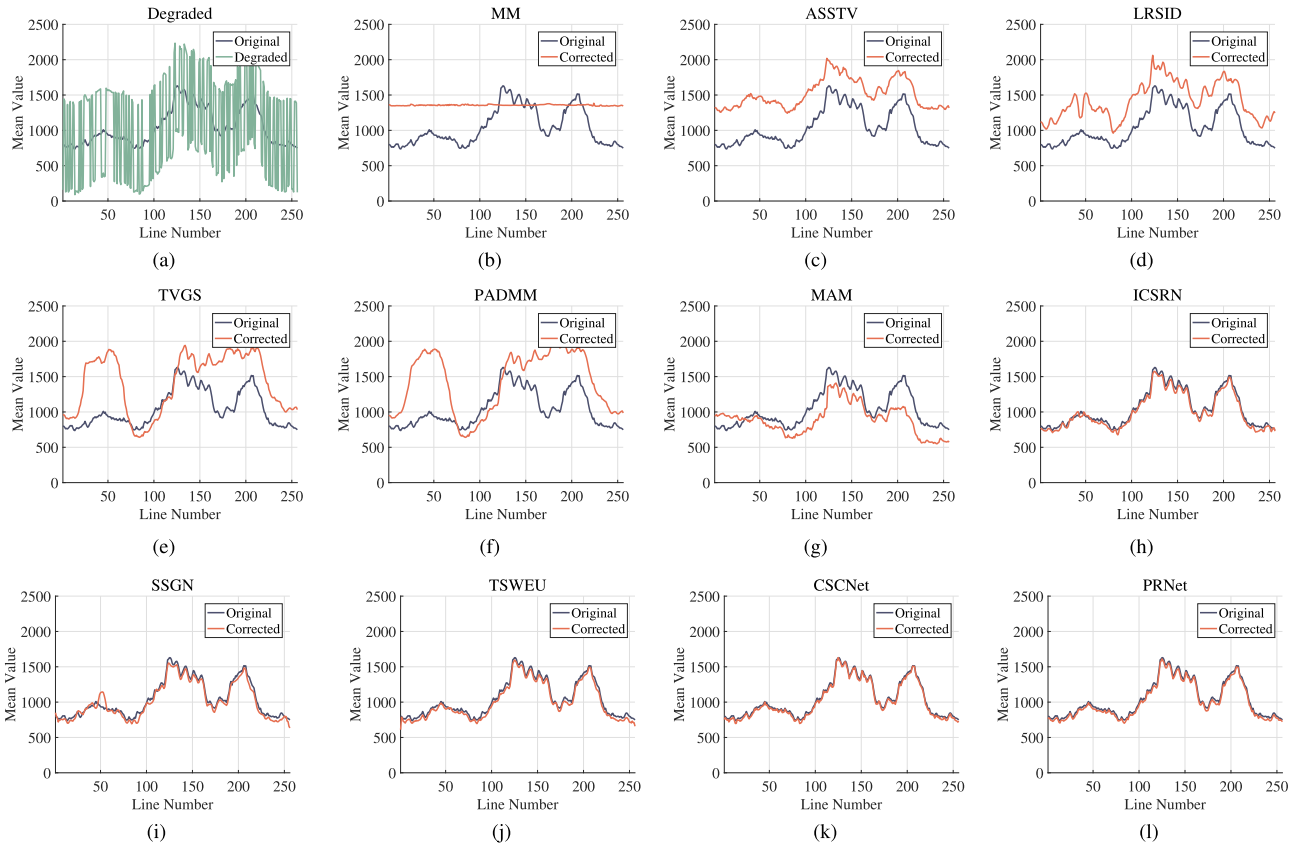


Fig. 14. Mean cross-profile analysis corresponding to Fig. 6 (PaviaU image for band 20). (a) Degraded image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) TVGS, (f) PADMM, (g) MAM, (h) ICSRN, (i) SSGN, (j) TSWEU, (k) CSCNet, and (l) PRNet.

TABLE IV
IMAGE NONUNIFORMITY USING TEN METHODS

Method	Band 85	Band 504
Original image	0.1016	0.1038
MM	0.0393	0.0333
ASSTV	0.0423	0.0671
LRSID	0.0495	0.0719
TVGS	0.0514	0.0597
PADMM	0.0383	0.0557
MAM	0.0372	0.0211
ICSRN	0.0525	0.0693
SSGN	0.0720	0.0829
TSWEU	0.0325	0.0653
CSCNet	0.0365	0.0680
PRNet	0.0291	0.0531

value of nonuniformity except for MM and MAM. However, as shown in Figs. 10(b) and 11(b), there is residual stripe noise in the restored images of MM. In addition, from Figs. 10(g) and (k) and 11(g) and (k), although the nonuniformity value of PRNet is slightly higher than MAM, the restored images of PRNet possess more detailed information. Compared with other destriping methods, the proposed PRNet considers the local spatial correlation, making the restored images more uniform.

D. Mean Cross-Profile Analysis

As shown in Figs. 14 and 15, the mean cross-profile of the restored results was analyzed to observe the strip removal effect more intuitively [33], where the abscissa means the line number (columns), and the ordinate is the mean value of the whole pixels located in the same column. Figs. 14 and 15 correspond to PaviaU image for band 20 and SWIR image for band 85, respectively. Specifically, the mean cross-profile curves of corrected images are expected to be similar to the original clean images for the simulated images (see Fig. 14). As for the real images, the mean cross-profile curves of corrected images should be smoother (see Fig. 15).

We can see from Fig. 14(a) that the original image information is severely damaged by the added stripe noise. MM adjusts the mean value of the whole columns to a unique value, which losses a lot of original image information. Corrected images by ASSTV and LRSID have a similar curve trend to the original image, while there is the certain gap between their values. The nonuniformity phenomenon between local regions caused by TVGS and PADMM in Fig. 6(f) and (g) is also reflected in Fig. 14(e) and (f). Generally speaking, the deep-learning-based methods provide the best corrected mean cross-profile curves compared with the traditional ones depending on the strong learning ability on data characteristics, including capturing the clean and stripe components. Among them, the corrected mean cross-profile curves generated by CSCNet and the proposed PRNet are almost completely consistent with the original clean images. From Fig. 10, most comparison methods cannot effectively remove the stripe noise in addition to MAM and PRNet, while the curve trend of the degraded image is damaged by MAM after the correction, as shown in Fig. 15(f). Compared with CSCNet, the corrected

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

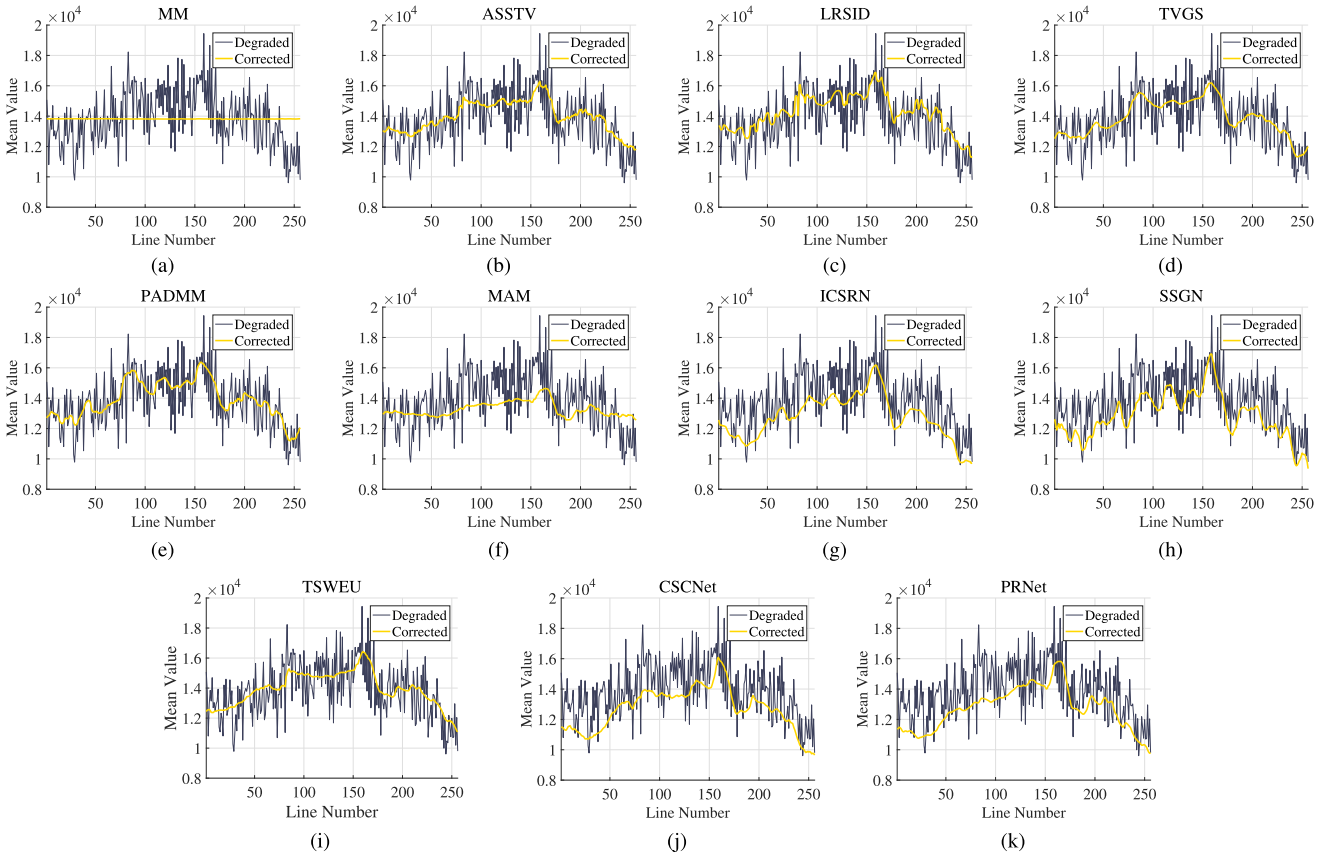


Fig. 15. Mean cross-profile analysis corresponding to Fig. 10 (SWIR image for band 85). Images corrected by (a) MM, (b) ASSTV, (c) LRSID, (d) TVGS, (e) PADMM, (f) MAM, (g) ICSRN, (h) SSGN, (i) TSWEU, (j) CSCNet, and (k) PRNet.

curve of PRNet is smoother. Overall, the proposed PRNet can accomplish the proper tradeoff between removing stripe noise and preserving image information.

In addition, to further evaluate the information loss on the destriping process, we present a visual assessment of the impact on detailed information at the sea–land boundary of the CHRIS_AM image for band 4. We compare the boundary regions of the original image with the clean image generated by PRNet in Fig. 16. The comparison demonstrates the effectiveness of PRNet in removing noise components while preserving detailed information to a significant extent. The proposed method maintains the clarity and integrity of local boundaries in the image, as well as protects more detailed features such as texture.

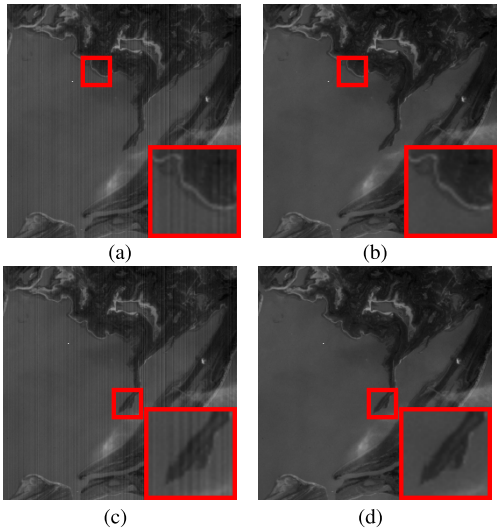


Fig. 16. Comparison of detailed information in CHRIS_AM images for band 4 at different positions. Detailed information of the original image at (a) position 1 and (c) position 2. Detailed information of the image corrected by PRNet at (b) position 1 and (d) position 2.

E. Model Complexity Analysis

To thoroughly analyze the balance between the effectiveness and efficiency of the destriping models, we also investigate the complexity of different models. Specifically, the network parameters of five deep-learning-based methods and inference time of all the methods are listed in Tables V and VI, respectively. From Table V, we can observe that ICSRN, SSGN, and PRNet are lightweight models. Since there are multiple parallel streams or correction modules in TSWEU and CSCNet to promote the destriping performance of models, their parameters are larger. Benefiting from the designed progressive destriping strategy, PRNet possesses small parameters and achieves promising restored results. From Table VI, it can be seen that the MM method and deep-learning-based methods process images with much speed compared with other methods, in which the inference time of PRNet is slightly higher than that of MM and SSGN.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

TABLE V
NETWORK PARAMETERS OF FIVE DEEP-LEARNING-BASED MODELS

Method	ICSRN	SSGN	TSWEU	CSCNet	PRNet
Parameters(M)	0.8	0.2	3.2	6.2	0.39

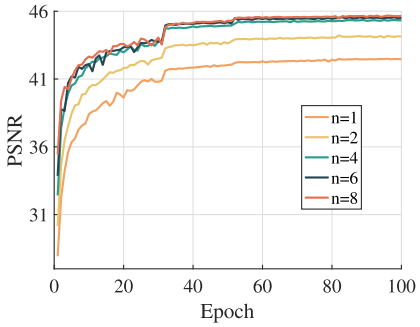


Fig. 17. PSNR comparison of the proposed PRNet with the different numbers of iterations during training. n stands for the number of iterations.

In general, PRNet achieves advanced destriping performance with relatively low model parameters and inference time.

F. Ablation Study

1) Iteration Number of MRM: PRNet removes the stripe noise multiple times in succession for one degraded image to deal with the different degrees of stripe noise. To investigate the effect of the iteration number of the MRM on the stripe removal performance, we examined the different numbers of iterations: 1, 2, 4, 6, and 8. The PSNR comparisons of different iterations are given in Fig. 17, and quantitative assessments on the PaviaU dataset are provided in Table VII. In Fig. 17, it can be seen that the learning ability of PRNet on the training set was strengthened as the iteration increases: that of values for four, six, and eight iterations are close to 46. From Table VII, four quantitative assessments are improved with the increase in the number of MRM iterations, while six and eight iterations possess similar performance. Therefore, we selected six iterations for PRNet by leveraging the correction performance, learning capacity, and network complexity.

In addition, we also investigate the influence of the progressive destriping strategy on the spectral information with different stripe noise levels. Fig. 18(a) and (b) shows the SAD and PSNR comparisons, respectively. From Fig. 18(a), we can observe that the SAD value tends to decrease and then increase with the number of iterations on the PaviaU dataset with three intensities of stripe noise and reaches a minimum value at six iterations. There is a certain degree of loss of spectral information in the destriping process, but choosing the appropriate number of iterations will ensure minimal loss of information. As can be seen from Fig. 18(b), PSNR appears the similar trends with SAD on the PaviaU dataset with Intensity-100 and Intensity-200. Although the PSNR values keep increasing on the PaviaU dataset with Intensity-300, its growth rate slowed down at six iterations. A comprehensive consideration of the degree of distortion of the spectral information and the quality of the restored image

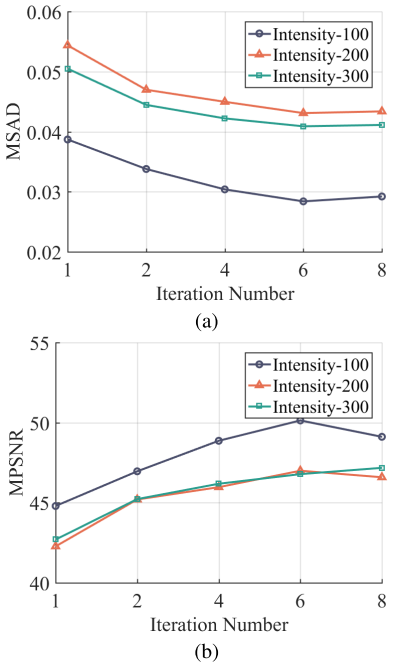


Fig. 18. (a) SAD and (b) PSNR comparisons of the proposed PRNet with different numbers of iterations on the PaviaU dataset with different intensities of stripe noise. Intensity-100, Intensity-200, and Intensity-300 indicate that the ranges of additive coefficients are $(-100, -50) \cup (50, 100)$, $(-200, -100) \cup (100, 200)$, and $(-300, -200) \cup (200, 300)$.

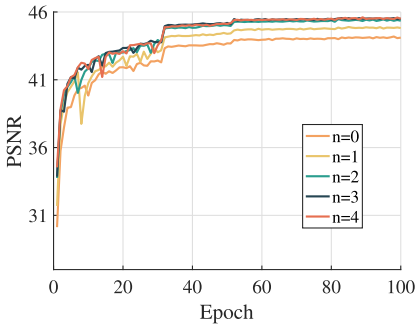


Fig. 19. PSNR comparison of the proposed PRNet with different numbers of PSN during training. n stands for the number of PSN.

led to the selection of six iterations for different intensities of stripe noise.

2) Number of PSN: In PRNet, the PSM is designed to exploit the local spatial correlation and further enhance the uniformity of the entire image, in which several PSNs are used to conduct patch sequences with different split scales. Thus, we implemented different numbers of PSN to assess the influence of the PSM on the destriping results. The corresponding PSNR comparisons during training and quantitative assessments are shown in Fig. 19 and Table VIII, respectively. When the number of PSN is 0, it means that PSM is completely removed from PRNet. From Fig. 19, we can see that the PSNR is the lowest when the number of PSN is 0, indicating the effectiveness of the PSM to promote the uniformity of the entire image. As the number of PSN increases, the performance of the proposed model on the training set was strengthened, while the PSNR values for two, three, and four PSNs are close to 46. Four quantitative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
INFERENCE TIME (IN SECONDS) OF EVERY METHOD ON THE CHRIS_AM IMAGE (FOR ONE BAND)

Method	MM	ASSTV	LRSID	TVGS	PADMM	MAM	ICSRN	SSGN	TSWEU	CSCNet	PRNet
Time	0.01	9.93	44.47	6.31	3.99	47.01	0.26	0.07	0.45	0.61	0.17

TABLE VII
QUANTITATIVE ASSESSMENTS OF PRNET WITH THE DIFFERENT NUMBERS OF ITERATIONS ON THE PAVIAU DATASET

Index	Iterations (1)	Iterations (2)	Iterations (4)	Iterations (6)	Iterations (8)
MSNR (↑)	29.7690	30.6952	31.0242	31.3422	31.3783
MPSNR (↑)	43.8339	44.7612	45.1772	45.4119	45.4486
MSSIM (↑)	0.9959	0.9964	0.9966	0.9969	0.9970
MSAD (↓)	0.0644	0.0633	0.0623	0.0623	0.0620

TABLE VIII
QUANTITATIVE ASSESSMENTS OF PRNET WITH THE DIFFERENT NUMBERS OF PSN ON THE PAVIAU DATASET

Index	PSN (0)	PSN (1)	PSN (2)	PSN (3)	PSN (4)
MSNR (↑)	30.1229	30.7918	31.1223	31.3422	31.0508
MPSNR (↑)	44.0905	44.8657	45.2137	45.4119	45.1318
MSSIM (↑)	0.9961	0.9966	0.9966	0.9969	0.9970
MSAD (↓)	0.0641	0.0626	0.0624	0.0623	0.0619

assessments listed in Table VIII demonstrate that more PSNs enhance the destriping results of PRNet. However, after adding four PSNs, the testing effect of model decreased slightly, which may be due to overfitting caused by the model being too complex. Therefore, we set three PSNs for PRNet.

V. CONCLUSION

In this article, we propose the PRNet for multispectral remote sensing image destriping. For better handling the different degrees of stripe noise, a progressive destriping strategy is designed to gradually eliminate the stripe noise, in which we present the MRM to iteratively process the results of stripe removal from the middle timesteps. Moreover, the PSM is introduced to explore the local spatial correlation to further strengthen the uniformity of the entire image. The effectiveness of the proposed model is verified by conducting extensive experiments on both the simulated and real remote sensing images. The superior visual performance and qualitative assessments of destriping demonstrate that the PRNet can effectively handle the stripe noise with different degrees to improve uniformity and preserve the structural information of the original image.

REFERENCES

[1] X. Jiang, M. Gong, T. Zhan, and M. Zhang, "Multiobjective endmember extraction based on bilinear mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8192–8210, Nov. 2020.

[2] S. Zhang et al., "Spectral–spatial hyperspectral unmixing using nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505713.

[3] C.-I. Chang, "Statistical detection theory approach to hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2057–2074, Apr. 2019.

[4] A. M. Saranathan and M. Parente, "Uniformity-based superpixel segmentation of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1419–1430, Mar. 2016.

[5] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[6] J. Lin, H. Zuo, Y. Ye, and X. Liao, "Histogram-based autoadaptive filter for destriping NDVI imagery acquired by UAV-loaded multispectral camera," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 648–652, Apr. 2019.

[7] N. Liu, W. Li, R. Tao, and J. E. Fowler, "Wavelet-domain low-rank/group-sparse destriping for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10310–10321, Dec. 2019.

[8] K. Naganuma, S. Takeyama, and S. Ono, "Zero-gradient constraints for destriping of remote-sensing data," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 1480–1484.

[9] K. Naganuma and S. Ono, "A general destriping framework for remote sensing images using flatness constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525016.

[10] J. Li, D. Zeng, J. Zhang, J. Han, and T. Mei, "Column-spatial correction network for remote sensing image destriping," *Remote Sens.*, vol. 14, no. 14, p. 3376, Jul. 2022.

[11] P. Xiao, Y. Guo, and P. Zhuang, "Removing stripe noise from infrared cloud images via deep convolutional networks," *IEEE Photon. J.*, vol. 10, no. 4, pp. 1–14, Aug. 2018.

[12] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, and L. Zhang, "Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7317–7329, Oct. 2019.

[13] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Advances in Neural Information Processing Systems*, vol. 28, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds. Red Hook, NY, USA: Curran Associates, Inc., 2015.

[14] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral–spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015.

[15] B. K. P. Horn and R. J. Woodham, "Destriping Landsat MSS images by histogram modification," *Comput. Graph. Image Process.*, vol. 10, no. 1, pp. 69–83, May 1979.

[16] M. Wegener, "Destriping multiple sensor imagery by improved histogram matching," *Int. J. Remote Sens.*, vol. 11, no. 5, pp. 859–875, May 1990.

[17] F. L. Gadallah, F. Csillag, and E. J. M. Smith, "Destriping multisensor imagery with moment matching," *Int. J. Remote Sens.*, vol. 21, no. 12, pp. 2505–2511, Jan. 2000.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.

[18] J. J. Simpson, J. R. Stitt, and D. M. Leath, "Improved finite impulse response filters for enhanced destriping of geostationary satellite data," *Remote Sens. Environ.*, vol. 66, no. 3, pp. 235–249, Dec. 1998.

[19] J. Chen, Y. Shao, H. Guo, W. Wang, and B. Zhu, "Destriping CMODIS data by power filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 9, pp. 2119–2124, Sep. 2003.

[20] R. Pande-Chhetri and A. Abd-Elrahman, "De-striping hyperspectral imagery using wavelet transform and adaptive frequency domain filtering," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 5, pp. 620–636, Sep. 2011.

[21] N. Acito, M. Diani, and G. Corsini, "Subspace-based striping noise reduction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 4, pp. 1325–1342, Apr. 2011.

[22] B. Münch, P. Trtik, F. Marone, and M. Stampanoni, "Stripe and ring artifact removal with combined Wavelet–Fourier filtering," *Opt. Exp.*, vol. 17, no. 10, pp. 8567–8591, 2009.

[23] P. Rakwatin, W. Takeuchi, and Y. Yasuoka, "Stripe noise reduction in MODIS data by combining histogram matching with facet filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1844–1856, Jun. 2007.

[24] H.-S. Jung, J.-S. Won, M.-H. Kang, and Y.-W. Lee, "Detection and restoration of defective lines in the SPOT 4 SWIR band," *IEEE Trans. Image Process.*, vol. 19, no. 8, pp. 2143–2156, Aug. 2010.

[25] J. Jia et al., "Destriping algorithms based on statistics and spatial filtering for visible-to-thermal infrared pushbroom hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4077–4091, Jun. 2019.

[26] H. Cui et al., "Multiscale intensity propagation to remove multiplicative stripe noise from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2308–2323, Apr. 2020.

[27] H. Shen and L. Zhang, "A MAP-based algorithm for destriping and inpainting of remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 5, pp. 1492–1502, May 2009.

[28] M. Bouali and S. Ladjal, "Toward optimal destriping of MODIS data using a unidirectional variational model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 8, pp. 2924–2935, Aug. 2011.

[29] Y. Chang, H. Fang, L. Yan, and H. Liu, "Robust destriping method with unidirectional total variation and framelet regularization," *Opt. Exp.*, vol. 21, no. 20, pp. 23307–23323, 2013.

[30] Y. Chang, L. Yan, H. Fang, and H. Liu, "Simultaneous destriping and denoising for remote sensing images with unidirectional total variation and sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 6, pp. 1051–1055, Jun. 2014.

[31] H. Carfantan and J. Idier, "Statistical linear destriping of satellite-based pushbroom-type images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 4, pp. 1860–1871, Apr. 2010.

[32] X. Liu, X. Lu, H. Shen, Q. Yuan, Y. Jiao, and L. Zhang, "Stripe noise separation and removal in remote sensing images by consideration of the global sparsity and local variational properties," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 3049–3060, May 2016.

[33] Y. Chang, M. Chen, L. Yan, X.-L. Zhao, Y. Li, and S. Zhong, "Toward universal stripe removal via wavelet-based deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2880–2897, Apr. 2020.

[34] X. Lu, Y. Wang, and Y. Yuan, "Graph-regularized low-rank representation for destriping of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4009–4018, Jul. 2013.

[35] J. Wright, A. Ganesh, S. Rao, Y. Peng, and Y. Ma, "Robust principal component analysis: Exact recovery of corrupted low-rank matrices via convex optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 22, 2009, pp. 1–12.

[36] Y. Chang, L. Yan, T. Wu, and S. Zhong, "Remote sensing image stripe noise removal: From image decomposition perspective," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7018–7031, Dec. 2016.

[37] Y. Chang, L. Yan, and S. Zhong, "Transformed low-rank model for line pattern noise removal," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1735–1743.

[38] Y.-Q. Zhao and J. Yang, "Hyperspectral image denoising via sparse representation and low-rank constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 296–308, Jan. 2015.

[39] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.

[40] Y. Chen, T.-Z. Huang, L.-J. Deng, X.-L. Zhao, and M. Wang, "Group sparsity based regularization model for remote sensing image stripe noise removal," *Neurocomputing*, vol. 267, pp. 95–106, Dec. 2017.

[41] W. Cao, Y. Chang, G. Han, and J. Li, "Destriping remote sensing image via low-rank approximation and nonlocal total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 848–852, Jun. 2018.

[42] Q. Xie et al., "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1692–1700.

[43] Y. Chen, T.-Z. Huang, and X.-L. Zhao, "Destriping of multispectral remote sensing image using low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4950–4967, Dec. 2018.

[44] H. Fan, C. Li, Y. Guo, G. Kuang, and J. Ma, "Spatial–spectral total variation regularized low-rank tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6196–6213, Oct. 2018.

[45] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5901–5909.

[46] J. Zhang, Q. Wu, C. Shen, J. Zhang, and J. Lu, "Multilabel image classification with regional latent semantic dependencies," *IEEE Trans. Multimedia*, vol. 20, no. 10, pp. 2801–2813, Oct. 2018.

[47] J. Zhang, Q. Wu, J. Zhang, C. Shen, and J. Lu, "Mind your neighbours: Image annotation with metadata neighbourhood graph co-attention networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2951–2959.

[48] H. Huang, J. Zhang, J. Zhang, J. Xu, and Q. Wu, "Low-rank pairwise alignment bilinear network for few-shot fine-grained image classification," *IEEE Trans. Multimedia*, vol. 23, pp. 1666–1680, 2021.

[49] J. Zhang, Z. Cai, F. Chen, and D. Zeng, "Hyperspectral image denoising via adversarial learning," *Remote Sens.*, vol. 14, no. 8, p. 1790, Apr. 2022.

[50] X. Kuang, X. Sui, Q. Chen, and G. Gu, "Single infrared image stripe noise removal using deep convolutional networks," *IEEE Photon. J.*, vol. 9, no. 4, pp. 1–13, Aug. 2017.

[51] Y. Zhong, W. Li, X. Wang, S. Jin, and L. Zhang, "Satellite-ground integrated destriping network: A new perspective for EO-1 hyperion and Chinese hyperspectral satellite datasets," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111416.

[52] J. Song, J.-H. Jeong, D.-S. Park, H.-H. Kim, D.-C. Seo, and J. C. Ye, "Unsupervised denoising for satellite imagery using wavelet directional CycleGAN," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6823–6839, Aug. 2021.

[53] Y. Chen, T.-Z. Huang, X.-L. Zhao, L.-J. Deng, and J. Huang, "Stripe noise removal of remote sensing images by total variation regularization and group sparsity constraint," *Remote Sens.*, vol. 9, no. 6, p. 559, Jun. 2017.

[54] H.-X. Dou, T.-Z. Huang, L.-J. Deng, X.-L. Zhao, and J. Huang, "Directional ℓ_0 sparse modeling for image stripe noise removal," *Remote Sens.*, vol. 10, no. 3, p. 361, Feb. 2018.

[55] J. Li, J. Zhang, F. Chen, K. Zhao, and D. Zeng, "Adaptive material matching for hyperspectral imagery destriping," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525220.

[56] Y. Wang et al., "New generation VNIR/SWIR/TIR airborne imaging spectrometer," *Proc. SPIE*, vol. 10156, pp. 203–210, Oct. 2016.



Jia Li received the B.E. degree from the Zhengzhou University of Light Industry, Zhengzhou, China, in 2019, and the Ph.D. degree in signal and information processing from Shanghai University, Shanghai, China, in 2023.

She is currently a Research Assistant with The First Affiliated Hospital, Zhengzhou University, Zhengzhou. Her research interests include computer vision and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.



Junjie Zhang received the bachelor's degree in computer science from the Nanjing University of Science and Technology, Nanjing, China, in 2014, and the Ph.D. degree in computer science from the University of Technology Sydney, Sydney, Australia, in 2019.

Then, he has worked as a Post-Doctoral Researcher with the University of Adelaide, Adelaide, Australia, until 2020. He is currently a Lecturer with Shanghai University, Shanghai, China. His research interests include computer vision, machine learning, and multimedia processing.



Jungong Han is currently the Chair Professor of computer vision with the Department of Computer Science, The University of Sheffield, Sheffield, U.K. He also holds an Honorary Professorship at The University of Warwick, Coventry, U.K. His research interests include computer vision, artificial intelligence, and machine learning.

Prof. Han is a fellow of the International Association of Pattern Recognition. He serves as an Associate Editor for several prestigious journals, such as IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and *Pattern Recognition*.



Chenggang Yan received the B.S. degree in computer science from Shandong University, Jinan, China, in 2008, and the Ph.D. degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2013.

He is currently a Professor with Hangzhou Dianzi University, Hangzhou, China. His research interests include intelligent information processing, machine learning, image processing, computational biology, and computational photography.



Dan Zeng (Senior Member, IEEE) received the B.S. degree in electronic science and technology and the Ph.D. degree in circuits and systems from the University of Science and Technology of China, Hefei, China, in 2008.

She is currently a Full Professor and the Dean of the Department of Communication Engineering, Shanghai University, Shanghai, China, directing the Computer Vision and Pattern Recognition Laboratory. Her main research interests include computer vision, multimedia analysis, and machine learning.

Dr. Zeng has been serving as a Technical Committee (TC) Member for IEEE Multimedia Systems and Applications (MSA), an Associate TC Member for IEEE Multimedia Signal Processing (MMSP), and an Associate Editor for the IEEE TRANSACTIONS ON MULTIMEDIA and the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:52:08 UTC from IEEE Xplore. Restrictions apply.